Chapter 15: Data Handling

Worksheet 15: Data Handling

Learning Objective: Collect, organise and summarise data using mean, median, mode and range, and construct and interpret bar graphs.

Worked Example:
Find the mean of the data: 12, 15, 18, 9, 16.
Solution: Mean $= \frac{12+15+18+9+16}{5} = \frac{70}{5} = 14$.

Questions (1–20)

1. Find the mean of: 4, 8, 6, 10, 12.

2. Find the mode of: 3, 5, 5, 7, 5, 9.

3. Find the median of: 7, 2, 9, 4, 5.

4. Find the range of: 18, 25, 12, 30, 20.

5. What does a bar graph use to represent data values?

6. The marks of 5 students in a test are 45, 60, 55, 70, 50. Find the mean mark.

7. Find the median of: 12, 18, 9, 22, 15, 7 (an even number of values).

8. A survey of 30 students' favourite colours gives: Blue-10, Red-8, Green-7, Yellow-5. Which colour is the mode?

9. Find the range of temperatures over a week: 28, 31, 25, 33, 27, 29, 30.

10. The weights (kg) of 5 friends are 32, 35, 30, 38, 35. Find the mode.

11. The runs scored by a cricketer in 6 matches are 45, 60, 0, 75, 30, 60. Find the mean, median and mode of his scores.

12. A bar graph shows books read by 5 students: A-12, B-8, C-15, D-10, E-9. Find the mean number of books read.

13. Find the median of: 23, 45, 12, 67, 34, 19, 50.

14. In a class test, scores were 78, 82, 78, 90, 85, 78, 92. Find the mode and explain what it tells you.

15. A company's monthly profits (in thousands of Rs.) over 6 months were 50, 65, 40, 70, 55, 60. Find the range and the mean profit.

16. The mean of 5 numbers is 20. If four of them are 18, 22, 25 and 15, find the fifth number.

17. A cricket team's mean score over 7 matches is 250 runs. In 6 matches they scored 280, 230, 260, 245, 255, 240. Find the score in the 7th match.

18. Student A's 5 test scores are 70, 75, 80, 85, 90 (mean 80). Student B's are 60, 70, 80, 90, 100 (mean 80). Both have the same mean – which student's scores are more consistent? Explain using the range.

19. Record the number of hours you spend on different activities (study, play, sleep) for one day, and represent the data as a table or bar graph.

20. Collect the heights (or ages) of 5 family members and find the mean, median, and mode of the data.

Reflection Question: Why do you think the mean, median and mode can sometimes give very different impressions of the “typical” value in a data set? Write your idea.

Real-Life Application Question: Find a real example of data in a newspaper, sports score table, or app (like average rainfall or average marks). Identify what kind of average (mean, median or mode) might have been used.